Data Detective

The mission where every graph turns into a claim you can check.

THE BIG QUESTION

Can a true average tell a false story?

HANDS-ON PROJECT

Ask 20 People

Write a real survey question, collect twenty real answers, and publish the findings with one honest limitation.

FLUENCY GAME

Predict the Roll

Call the outcome in writing, tally fifty trials, then explain the gap between prediction and result.

AXIOM · MISSION COMMANDER, NINE TOURS

“Brock — every briefing arrives with an average on it. Your job is to ask what got averaged.”

MISSION 07 · BEFORE WE LAUNCH

Pre-Assessment

Twelve items, six strands. Anything you already own, we skip.

NAME ______________

DATE ______________

This is not a test. Nothing here counts for a grade. Two of these ask you to explain rather than calculate — write the sentence anyway.

STRAND A
READING GRAPHS

A1 On a pictograph each ★ = 5 books. A row has 6 ★. How many books?

A2 A bar chart is marked every 10. A bar stops halfway between 20 and 30. Its value?

STRAND B
MAKING A GRAPH

B1 Your biggest value is 46. What scale would you mark the axis in?

B2 Name the two things every graph must have or it proves nothing.

STRAND C
MEAN & MEDIAN

C1 Mean of 4, 6 and 8?

C2 Median of 9, 2, 5?

STRAND D
MODE & RANGE

D1 Mode of 5, 7, 7, 9?

D2 Range of 3, 9, 12?

STRAND E
OUTLIERS

E1 Mean of 5, 5, 5, 25?

E2 Median of the same four numbers. Why is it so different?

STRAND F
CHANCE

F1 How many ways can one die land on an even number?

F2 A bag holds 3 red and 5 blue. Is drawing red likely or unlikely?

TEACHER PAGE · DO NOT HAND TO STUDENT

Skip Chart

Score the pre-assessment, then cross days off the three-week calendar.

2 / 2 → SKIP
1 / 2 → COMPRESS
0 / 2 → TEACH

STRAND	ITEMS	COVERS	SCORE	IF 2/2, REMOVE
A · Reading graphs	A1 A2	Week 1 · lesson 1.1	___ / 2	Day 1.1. Warm-Up tier drops to 3 items all unit.
B · Making a graph	B1 B2	Week 1 · lesson 1.2	___ / 2	The Warm-Up half of 1.2 only. He still draws his own survey graph in Week 2.
C · Mean & median	C1 C2	Week 1 · lesson 1.3	___ / 2	Nothing. Keep 1.3 whatever he scores — the Big Question is built on it.
D · Mode & range	D1 D2	Week 1 · lesson 1.3 Warm-Up	___ / 2	The Warm-Up row on 1.3. Go straight to the Core.
E · Outliers	E1 E2	Week 1 · lesson 1.4	___ / 2	Nothing — but if he explains E2 well, move 1.4 to Challenge only and spend Thursday on Week 3's misleading graph.
F · Chance	F1 F2	Friday enrichment · Week 2 lesson 2.4	___ / 2	Nothing. Predict the Roll stays — it is fifty rolls and it teaches the gap between theory and result.

ONE NOTE BEFORE YOU START

This mission is three weeks, not five, because the project does the teaching. If the survey stalls — nobody home, nobody willing — let him survey the same twenty people about something else rather than lose the week. The maths is in the analysis, not the topic.

MISSION 07 · WEEK 1 · MONDAY · LESSON 1.1

NAME ___________

DATE ___________

Reading a Graph

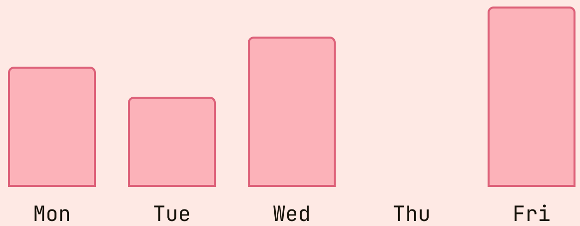
Find the scale first. Everything else depends on it.

✦ Warm-Up

EACH ★ = 5 BOOKS

- 3 ★ = ? books
- 6 ★ = ? books
- 40 books = ? ★
- half a ★ = ? books
- 2½ ★ = ? books
- 90 books = ? ★

BOOKS READ THIS WEEK · THE GRAPH YOU'LL BE ASKED ABOUT



The axis is marked every 3 pages. Mon 12, Tue 9, Wed 15, Thu 0, Fri 18. Thursday counts as a day even though the bar is empty — leaving it out is the commonest way to make data lie.

◆ Core

USE THE GRAPH ABOVE

1. Pages read across the whole week
2. How many more on Friday than on Tuesday
3. The mean number of pages per day, counting Thursday
4. The mean if you leave Thursday out altogether
5. Which of those two means would you put in a report, and why?

★ Challenge

NOT GRADED

C1. A pictograph has four rows and each ★ = 5. The rows total 90 books. How many stars are drawn altogether?

C2. Redraw Monday to Friday as a pictograph where each ★ = 6 pages. Which days need a part-star, and how would you draw it?

AXIOM: Read the scale out loud before you read a single bar. Half the wrong answers in this mission come from assuming one square means one thing.

NAME ___________

DATE ___________

Building a Graph

Tally, choose a scale, draw the bars, label both axes.

Warm-Up

TURN EACH TALLY INTO A NUMBER

- |||| ______
- two fives ______
- three fives + 2 ______
- four fives + 3 ______
- 20 as fives ______
- 17 as fives + ? ______

THE DATA · FAVOURITE FRUIT, 30 PEOPLE ASKED

apple	**12**
banana	**8**
grape	**6**
pear	**4**

Draw this as a bar graph in the grid below. Pick a scale that fits: going up in ones needs twelve squares, going up in twos needs six. Label the vertical axis with your scale and the horizontal axis with the fruit.

Core

1. People asked altogether ....
2. The mode of this data ....
3. Apple votes minus pear votes ....
4. Mean votes per fruit ....
5. Out of 100 people, how many would you expect to pick apple? Show the reasoning in one line.

....

Challenge

C1. Someone draws this graph with the vertical axis starting at 4 instead of 0. What happens to the pear bar, and is the graph still honest?

....

C2. Only 30 people were asked. Write one sentence saying what this graph *cannot* prove.

....

NAME ___________

DATE ___________

The Four Measures

Mean, median, mode, range — on one set of numbers, side by side.

Warm-Up

SMALL SETS, WHOLE ANSWERS

mean of 4, 6, 8	median of 2, 5, 9	mode of 5, 7, 7, 9	range of 3, 9, 12	mean of 10 and 20	total of 6, 7, 8, 9

THE METHOD, ONCE — THEN USE IT ALL WEEK

Mean
Add them all up, divide by how many. It levels the set off.

Median
Put them in order. Take the middle one. Two middles? Halfway between.

Mode
The value that shows up most often. There can be none, or several.

Range
Biggest take smallest. It measures spread, not the middle.

Core

THE SET: 4, 7, 7, 9, 13

1. Their total ....

2. The mean ....

3. The median ....

4. The mode and the range ....

5. Five numbers have a mean of 10. Four of them are 6, 8, 12 and 14. Find the fifth.

Challenge

C1. Invent five whole numbers whose mean is 8 and whose median is 6. Prove both.

C2. Can a set have a mean that is not one of its own numbers? Give an example, then say whether the same is true of the median.

MISSION 07 · WEEK 1 · THURSDAY · LESSON 1.4

NAME ___________

DATE ___________

When the Average Lies

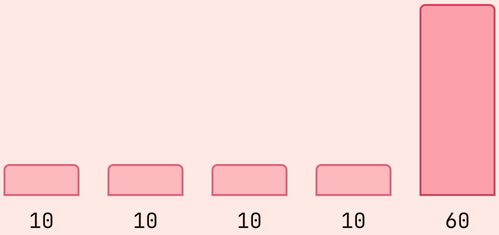
One extreme value moves the mean and leaves the median alone.

Warm-Up

TWO SETS, FOUR QUESTIONS

mean of 5, 5, 5	median of 5, 5, 5, 5	mean of 5, 5, 5, 25	median of 5, 5, 5, 25	range of 5, 5, 5, 25	mode of 5, 5, 5, 25

FIVE PEOPLE IN A ROOM · POCKET MONEY PER WEEK



Four people get 10 a week. One gets 60. The mean of this room is 20 — a figure that describes not one single person standing in it. The median is 10, and four out of five people are on exactly that.

Core

USE THE ROOM ABOVE

1. Mean of the five ....
2. Median of the five ....
3. Range of the five ....
4. Mean once the 60 walks out ....
5. A headline says “average pocket money in this room: 20 a week.” It is true. Write one sentence explaining why it is also misleading.

Challenge

C1. Six houses on a street cost 200 each and one costs 900. Find the mean and the median. Which one would an estate agent print, and which one would a buyer want?

C2. Build a set of seven numbers where the mean is more than double the median. Then say in one line what you had to do to get it.

NAME ___________

DATE ___________

Predict the Roll

Write the prediction first. A prediction you can change afterwards is not one.

You need: one die and a pencil. Fifty rolls takes about six minutes. Nobody wins this game — the point is the gap between what should happen and what did.

Step 1 · Predict, in ink

Fifty rolls, six faces. How many times should each face appear?

....

Fifty does not divide by six. How many rolls are left over?

....

Step 2 · Roll fifty times and tally

FACE	TALLY	COUNT	PREDICTED	DIFFERENCE
1				
2				
3				
4				
5				
6				

Step 3 · Explain the gap

C1. Which face missed its prediction by the most? Is the die broken?

....

C2. If you rolled 600 times instead of 50, would the counts get closer to the prediction or further away? Say why.

....

C3. Two dice, added. Which total should come up most often, and how many ways can it happen?

....

Weeks 2–3

Outlined, not written out. Week 1's five worksheets are the model — build these from the app's practice sets.

QUIZ · end of week 2
TEST · end of week 3

Week 2 · Survey & Probability QUIZ FRIDAY

He writes his own question, collects twenty real answers, and meets the difference between likely and actual.

MON 2.1
Write the question
Rewrite three loaded questions until they stop pushing an answer.

TUE 2.2
Ask 20 people
Tallied on paper as they answer. No tidying up afterwards.

WED 2.3
Graph it
Bar graph of his own data, plus all four measures.

THU 2.4
Chance as a number
Impossible to certain, written as a count out of the total.

FRI
Mid-unit quiz
Page 10. Eight items, 85% to keep flying.

Week 3 · Proof & Project UNIT TEST

Ask 20 People gets published — a headline, a graph, and one sentence about what twenty people cannot prove.

MON 3.1
Publish it
One headline, one graph, one honest limitation.

TUE 3.2
The misleading graph
Start the axis at 90 and watch a tiny gap look enormous.

WED 3.3
50 trials
Predict, roll, tally, and explain the gap in writing.

THU
Error journal sweep
Re-read every entry. Fix only what repeats.

FRI
Mission 07 test
Pages 11–12. Twelve items plus one explanation.

IF THE SURVEY GOES WRONG

Fewer than twenty answers is fine — twelve is enough to compute everything. What is not fine is inventing the missing eight. If he wants to, that is the lesson: made-up data has no error to find.

WATCH FOR

Computing the median without sorting first. It gives the right answer often enough to feel safe, which is exactly why it is worth catching this week rather than next year.

NAME ______________

SCORE ______ / 8

Mid-Unit Quiz

Eight items from Weeks 1 and 2. 85% to keep flying — that's 7 out of 8.

1. Each ★ = 5 books. A row has 7 ★. How many books? ....

2. Find the mean of 6, 8, 10 and 12. ....

3. Find the median of 11, 4, 7, 20, 9. ....

4. Find the mode and the range of 3, 8, 8, 8, 15. ....

5. Six numbers have a mean of 9. What do they total? ....

6. A bag holds 4 red and 6 blue. How many draws out of 10 should be red? ....

7. In 60 rolls of one die, how many 5s would you expect? ....

8. A set is 2, 2, 2, 2 and 22. Give the mean and the median, then say in one sentence which one describes a typical member of the set.

....

....

Marking note: item 8 is worth the same as any other. A right mean with no sentence is half marks — the sentence is the skill this mission exists for.

ANSWERS ON PAGE 15

Unit Test · Part 1

NAME ______________

DATE ______________

Items 1–8. Take as long as you need; there is no time limit on this test.

1. A bar chart marked every 4 has a bar at the third mark. Its value? ....

2. Each ▮ = 2 goals. A team's bar is 9 ▮ long. Goals? ....

3. Mean of 14, 16, 18 and 20. ....

4. Median of 5, 12, 3, 8, 21, 9. ....

5. Range of 7, 7, 19, 4, 11. ....

6. Five scores total 45. A sixth score of 9 is added. New mean? ....

7. A spinner has 8 equal parts, 2 of them gold. In 40 spins, expected golds? ....

8. Two dice are rolled and added. How many ways can the total be 7? ....

CONTINUES ON PAGE 12

SCORE ______ / 12

BAND ______________

Unit Test · Part 2

Items 9–12 and the Big Question.

9. Twenty people were asked their favourite drink: water 8, juice 6, milk 4, tea 2. Draw the bar graph, labelled, with your scale stated.

10. From that survey: how many out of 100 people would you expect to pick water? ....

11. The mode of that survey. ....

12. Mean number of votes per drink. ....

EXPLAIN YOUR THINKING · THE BIG QUESTION

Can a true average tell a false story?

Use one set of numbers you invent. Give its mean and its median, then say which one you would print and who it would mislead. Scored on reasoning, not neatness.

....

....

....

....

....

....

11–12
Gold band. Badge earned.

9–10
Silver band. One reteach day, then the badge.

≤ 8
Bronze. Reteach the weak strand and retest that part only.

Key A

Pre-assessment, Lesson 1.1 and Lesson 1.2.

Mark the reasoning, not the handwriting. Any correct method counts.

Pre-assessment

A1 · 30 | A2 · 25 | B1 · in 5s or 10s (either, if justified) | B2 · labelled axes and a stated scale

C1 · 6 | C2 · 5 | D1 · 7 | D2 · 9 | E1 · 10 | E2 · 5; the 25 pulls the mean but not the middle

F1 · 3 | F2 · unlikely — 3 out of 8 is less than half

Lesson 1.1 · Reading a Graph

Warm-Up · 15 | 30 | 8 | 2½ | 12½ | 18

Core · 1 · 54 | 2 · 9 | 3 · 10.8 | 4 · 13.5 | 5 · the one counting Thursday, because Thursday happened

C1 · 18 stars | C2 · Mon 2 ★, Tue 1½ ★, Wed 2½ ★, Thu none, Fri 3 ★ — a part-star is drawn as a fraction of the symbol

Warm-Up items 4 and 5 are the only fractional answers in this mission; accept 2.5 and 12.5 written either way.

Lesson 1.2 · Building a Graph

Warm-Up · 4 | 10 | 17 | 23 | 4 fives | 3 fives + 2

Core · 1 · 30 | 2 · apple | 3 · 8 | 4 · 7.5 | 5 · 40 (12 out of 30 is 4 out of 10)

C1 · the pear bar disappears and the apple bar looks eight times the size of nothing — the graph is dishonest even though every number on it is right. C2 · it cannot prove what everyone likes, only what these thirty said.

Key B

Lesson 1.3, Lesson 1.4 and Friday's enrichment.

Where a question asks for a sentence, no sentence means no mark — even with the number right.

Lesson 1.3 · The Four Measures

Warm-Up · 6 | 5 | 7 | 9 | 15 | 30

Core · 1 · 40 | 2 · 8 | 3 · 7 | 4 · mode 7, range 9 | 5 · 10 (the five must total 50)

C1 · many answers; 2, 4, 6, 8, 20 works — mean 8, median 6. C2 · yes, the mean of 4 and 6 is 5, which is in neither; the median of an odd-sized set is always one of its own numbers, but the median of an even-sized set need not be.

Lesson 1.4 · When the Average Lies

Warm-Up · 5 | 5 | 10 | 5 | 20 | 5

Core · 1 · 20 | 2 · 10 | 3 · 50 | 4 · 10 | 5 · sentence

Item 5 · full marks for any sentence saying the 60 drags the mean upward so that four of the five people are below it. C1 · mean 300, median 200; the agent prints the mean, the buyer wants the median. C2 · any set with one very large value, e.g. 1, 1, 1, 2, 2, 2, 70.

Friday · Predict the Roll

Step 1 · about 8 each | 2 rolls left over Step 2 · his own tally; the counts must total 50

C3 · 7 is the most likely total, and it can happen 6 ways

C1 · no — a spread of a few either way is normal in fifty rolls. C2 · closer, because more trials give the underlying chance more room to show. If he says “further away” because the raw difference grows, that is worth discussing rather than marking wrong.

TEACHER PAGE · ANSWER KEY

Key C

Mid-unit quiz, unit test, and the error journal sheet.

Quiz is out of 8, test out of 12. 85% is 7/8 and 11/12.

Mid-unit quiz

1 · 35

2 · 9

3 · 9

4 · mode 8, range 12

5 · 54

6 · 4

7 · 10

8 · mean 6, median 2 — the median

Unit test

1 · 12

2 · 18

3 · 17

4 · 8.5

5 · 15

6 · 9

7 · 10

8 · 6 ways

10 · 40

11 · water

12 · 5

Item 9 and the Big Question

Item 9 earns its mark for a stated scale and two labelled axes, not for straight bars. The Big Question is full marks for any invented set where the mean and median differ, both computed correctly, plus a named audience who would be misled. A perfect answer does not need the word "outlier" — it needs the idea.

Error journal · Mission 07

ONE LINE PER MISTAKE · STUDENT PAGE

Write what went wrong, not what the right answer was. On Thursday of Week 3, read the whole list and fix only what appears twice.

DAY	WHAT I GOT WRONG, AND WHY